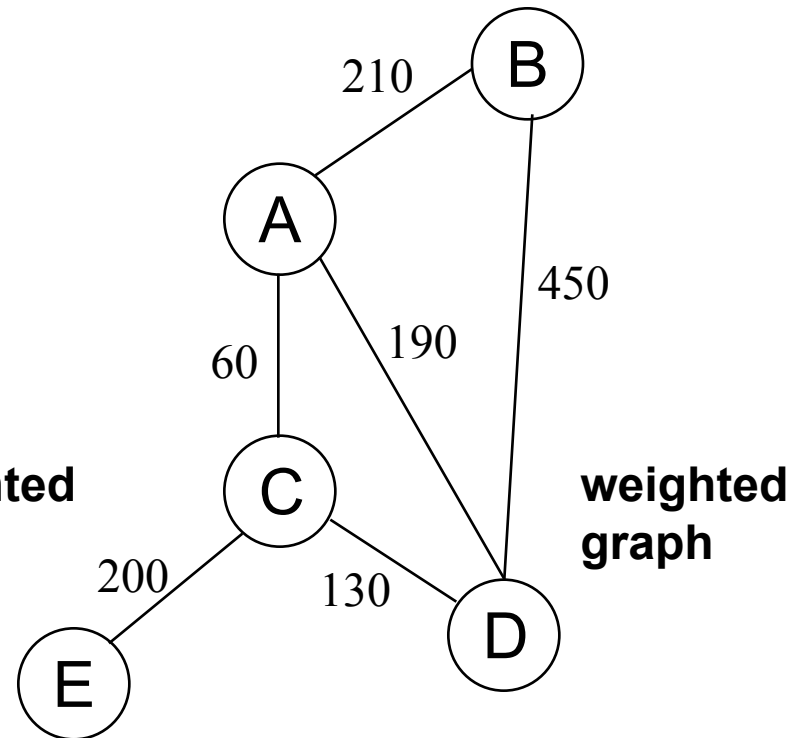
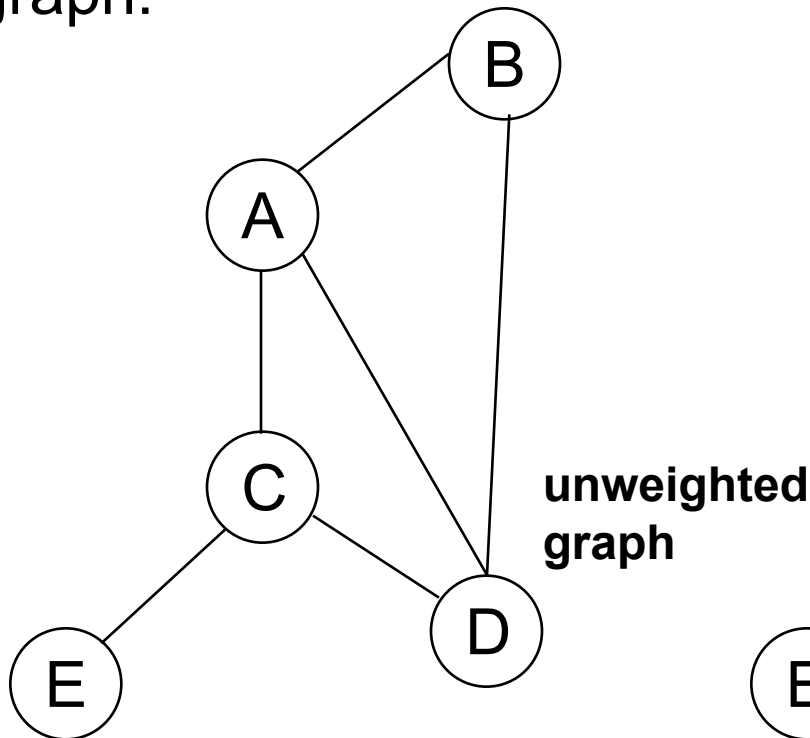


Shortest Path

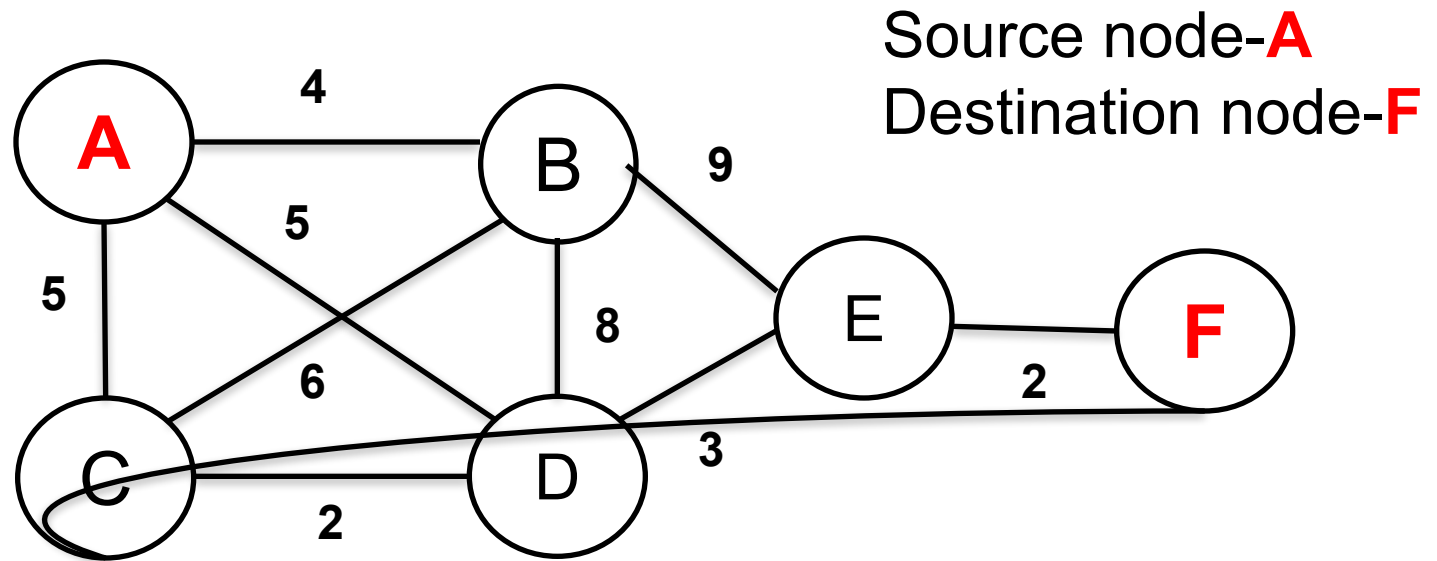
Single Source Shortest Path
(Dijkstra's Algorithm)

Shortest Path Problems

- **What is shortest path ?**
 - shortest length between two vertices for an unweighted graph:
 - smallest cost between two vertices for a weighted graph:



Example



Steps:

10

- Find the shortest path between two vertices in a graph
- Calculate all vertices from the graph

10

A-D-E-F = 10

Shortest Path Problems

- How can we find the shortest route between two points on a map?
- Model the problem as a graph problem:
 - Road map is a weighted graph:
 - vertices** = cities
 - edges** = road segments between cities
 - edge weights** = road distances
 - Goal: find a shortest path between two vertices (cities)

Shortest Path Problems

- **Input:**

- Directed graph $G = (V, E)$
- Weight function $w : E \rightarrow \mathbf{R}$

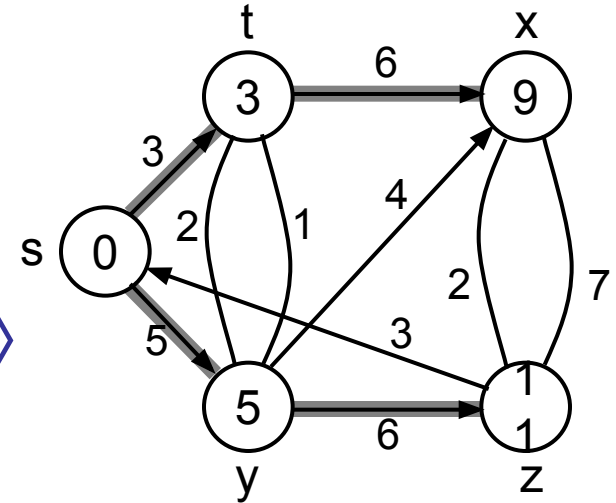
- **Weight of path** $p = \langle v_0, v_1, \dots, v_k \rangle$

$$w(p) = \sum_{i=1}^k w(v_{i-1}, v_i)$$

- **Shortest-path weight** from u to v :

$$\delta(u, v) = \begin{cases} \min w(p) : u \xrightarrow{p} v & \text{if there exists a path from } u \text{ to } v \\ \infty & \text{otherwise} \end{cases}$$

- Shortest path u to v is any path p such that $w(p) = \delta(u, v)$



Variants of Shortest Paths

- **Single-source shortest path**

- $G = (V, E) \Rightarrow$ find a shortest path from a given source vertex s to each vertex $v \in V$

- **Single-destination shortest path**

- Find a shortest path to a given destination vertex t from each vertex v
- Reverse the direction of each edge $\Rightarrow$ single-source

- **Single-pair shortest path**

- Find a shortest path from u to v for given vertices u and v
- Solve the single-source problem

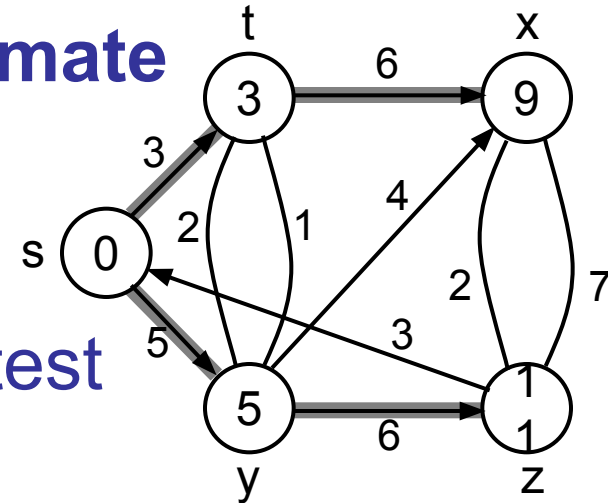
- **All-pairs shortest-paths**

- Find a shortest path from u to v for every pair of vertices u and v

Shortest-Path Representation

For each vertex $v \in V$:

- $d[v] = \delta(s, v)$: a **shortest-path estimate**
 - Initially, $d[v] = \infty$
 - Reduces as algorithms progress
- $\pi[v]$ = **predecessor** of v on a shortest path from s
 - If no predecessor, $\pi[v] = \text{NIL}$
 - π induces a tree—**shortest-path tree**
- Shortest paths & shortest path trees are not unique



Initialization

Alg.: INITIALIZE-SINGLE-SOURCE(V, s)

1. **for** each $v \in V$
2. **do** $d[v] \leftarrow \infty$
3. $\pi[v] \leftarrow \text{NIL}$
4. $d[s] \leftarrow 0$

- All the shortest-paths algorithms start with INITIALIZE-SINGLE-SOURCE

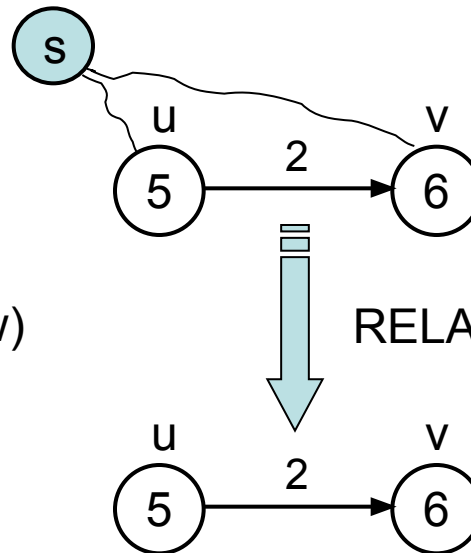
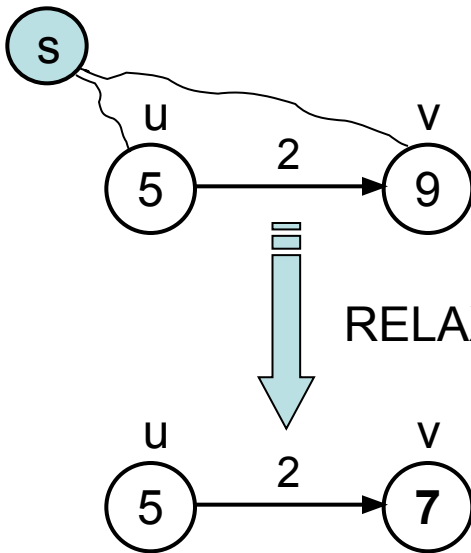
Relaxation

- **Relaxing** an edge (u, v) = testing whether we can improve the shortest path to v found so far by going through u

If $d[v] > d[u] + w(u, v)$

we can improve the shortest path to v

$\Rightarrow$ update $d[v]$ and $\pi[v]$



After relaxation:
 $d[v] \leq d[u] + w(u, v)$

RELAX(u, v, w)

1. if $d[v] > d[u] + w(u, v)$
2. then $d[v] \leftarrow d[u] + w(u, v)$
3. $\pi[v] \leftarrow u$

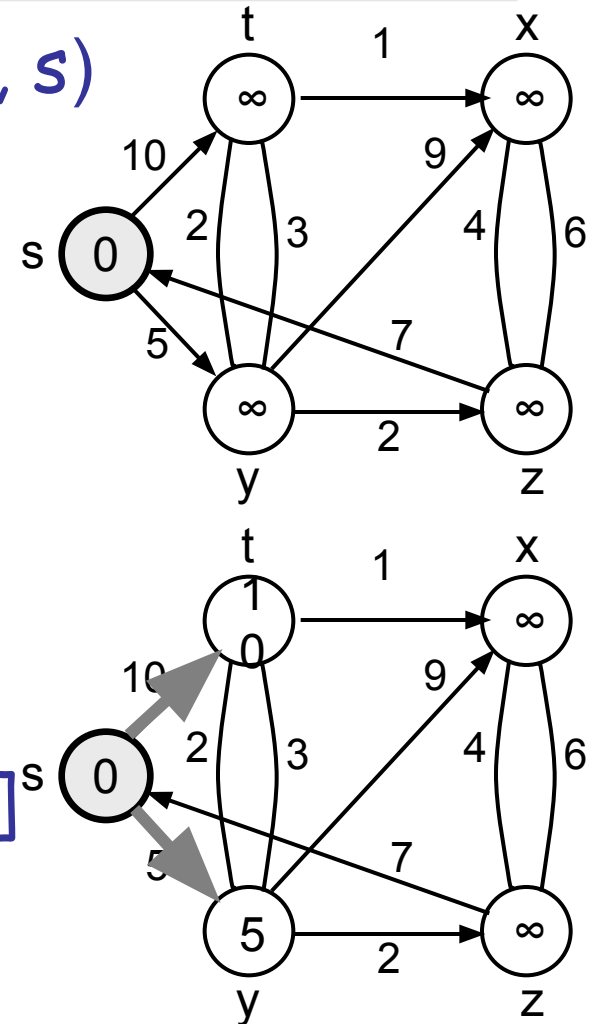
- All the single-source shortest-paths algorithms
 - start by calling INIT-SINGLE-SOURCE
 - then relax edges
- The algorithms differ in the order and how many times they relax each edge

Dijkstra's Algorithm

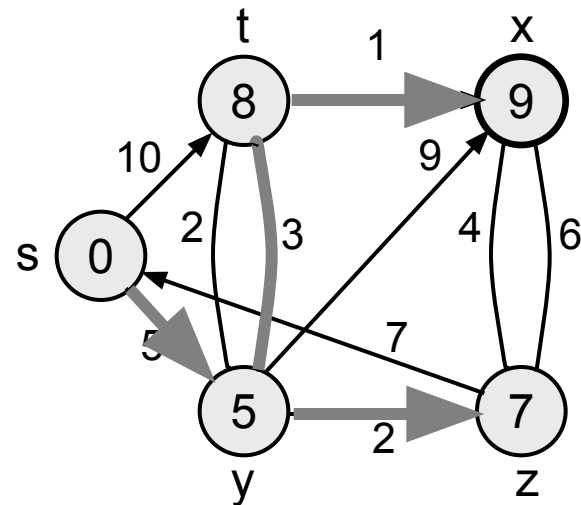
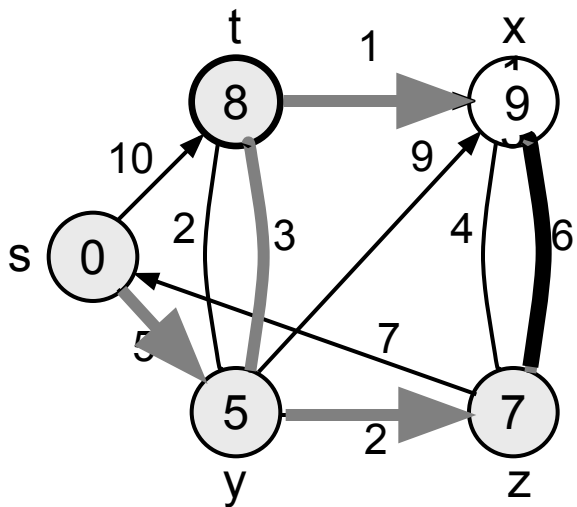
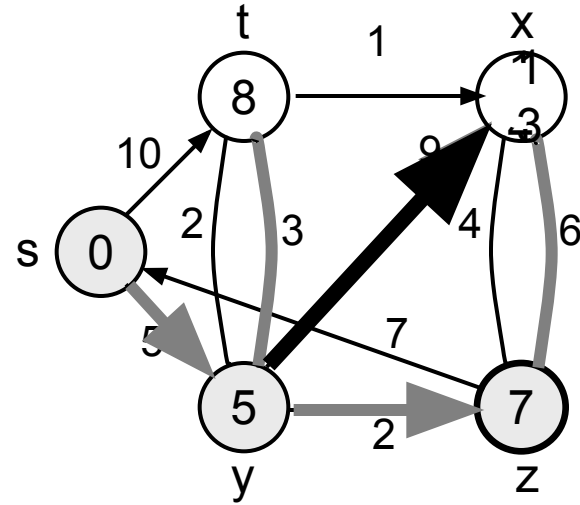
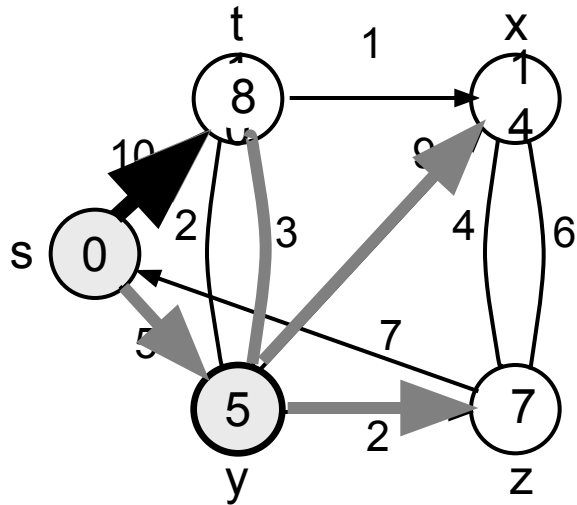
- Single-source shortest path problem:
 - No negative-weight edges: $w(u, v) > 0 \quad \forall (u, v) \in E$
- Maintains two sets of vertices:
 - S = vertices whose final shortest-path weights have already been determined
 - Q = vertices in $V - S$: min-priority queue
 - Keys in Q are estimates of shortest-path weights ($d[v]$)
- Repeatedly select a vertex $u \in V - S$, with the minimum shortest-path estimate $d[v]$

Dijkstra (G, w, s)

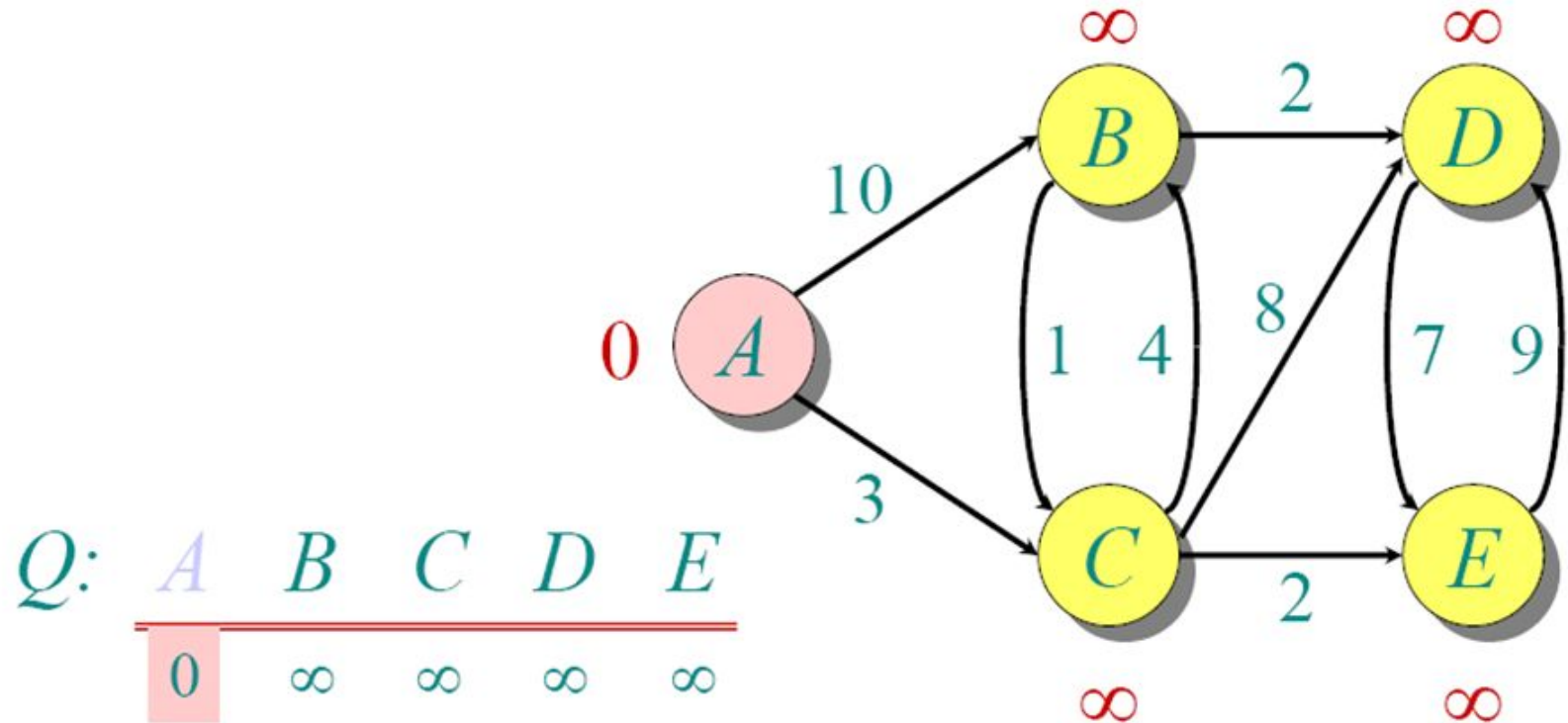
1. INITIALIZE-SINGLE-SOURCE(V, s)
2. $S \leftarrow \emptyset$
3. $Q \leftarrow V[G]$
4. **while** $Q \neq \emptyset$
5. **do** $u \leftarrow \text{EXTRACT-MIN}(Q)$
6. $S \leftarrow S \cup \{u\}$
7. **for** each vertex $v \in \text{Adj}[u]$
8. **do** RELAX(u, v, w)



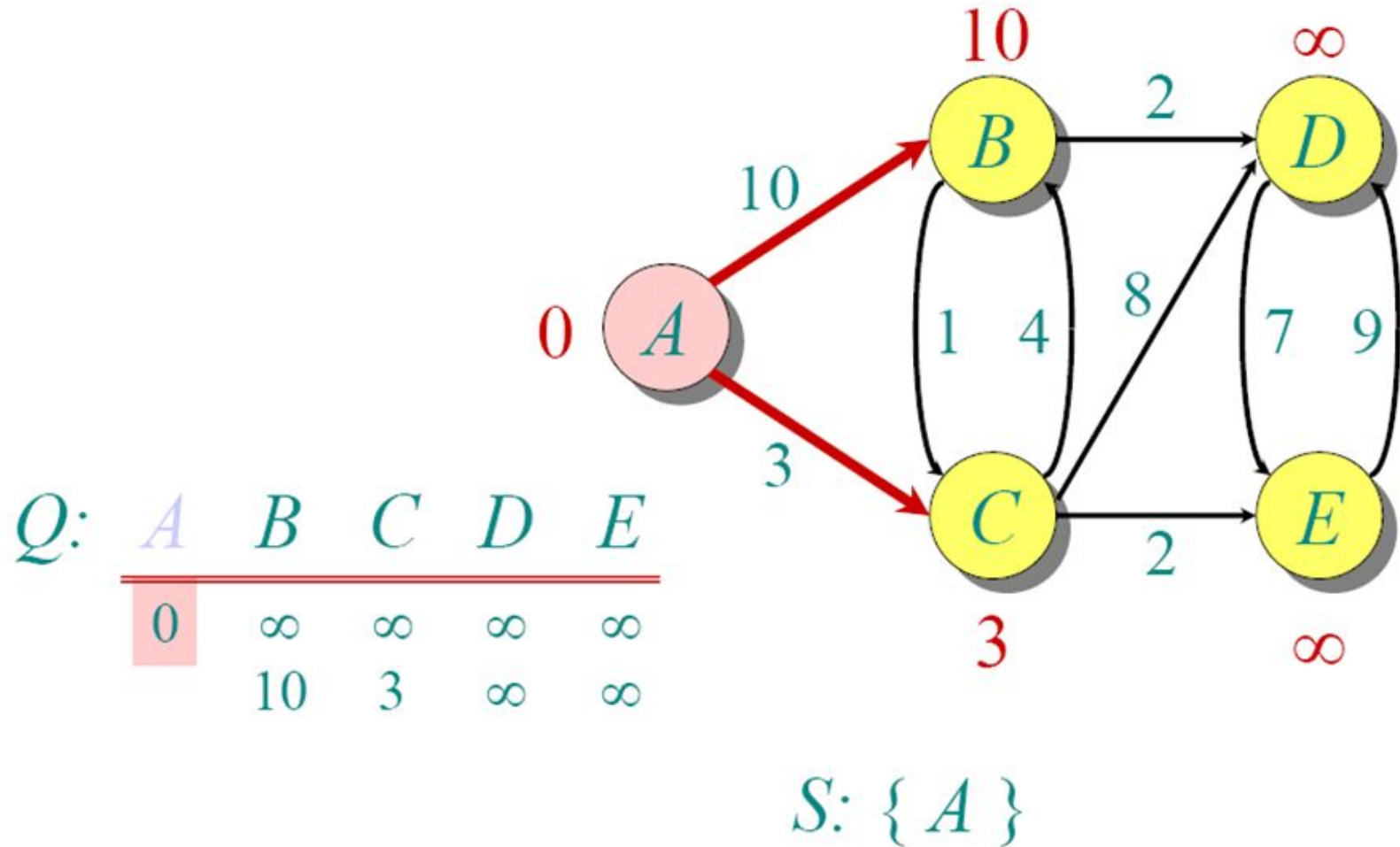
Example



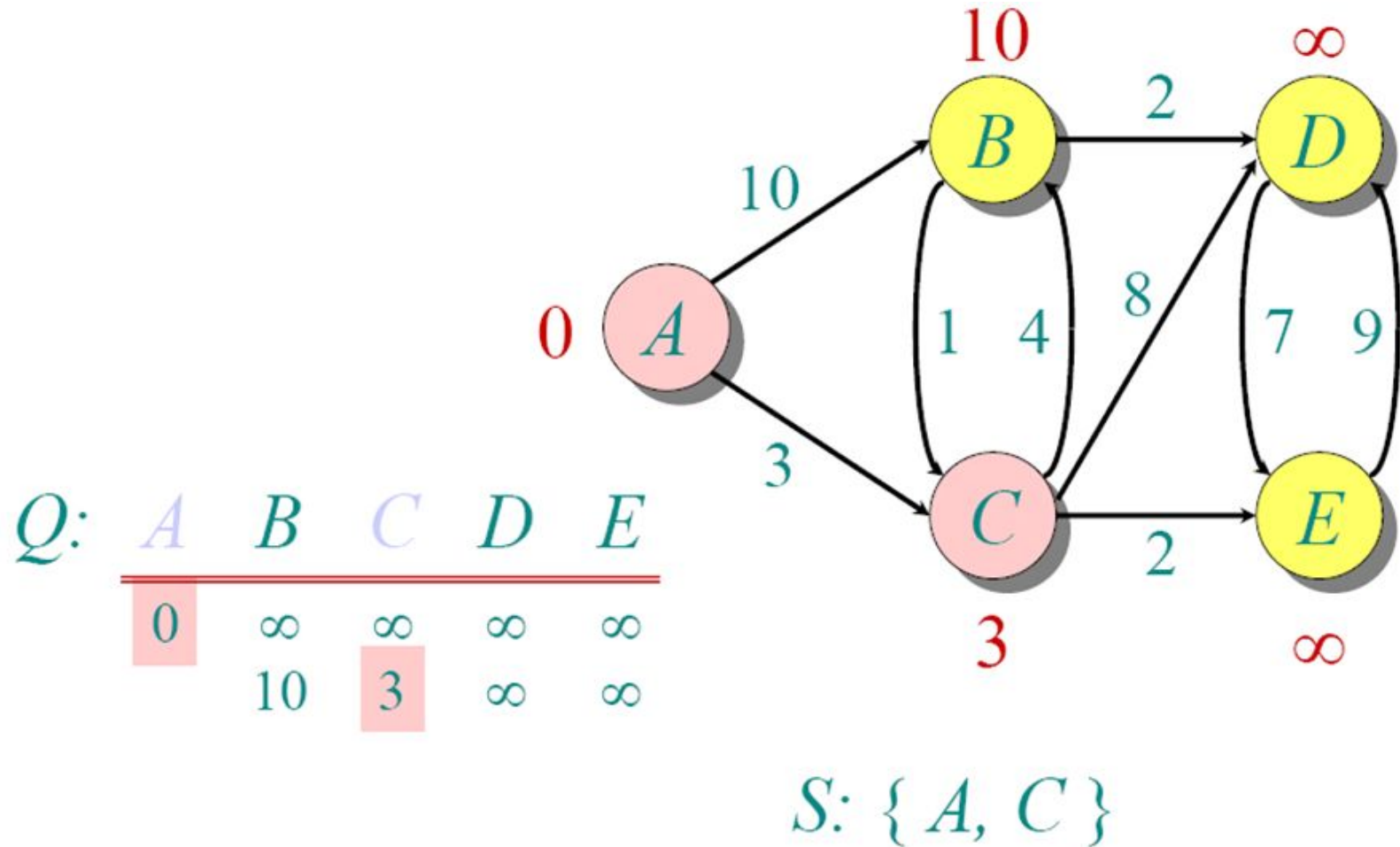
Another Example



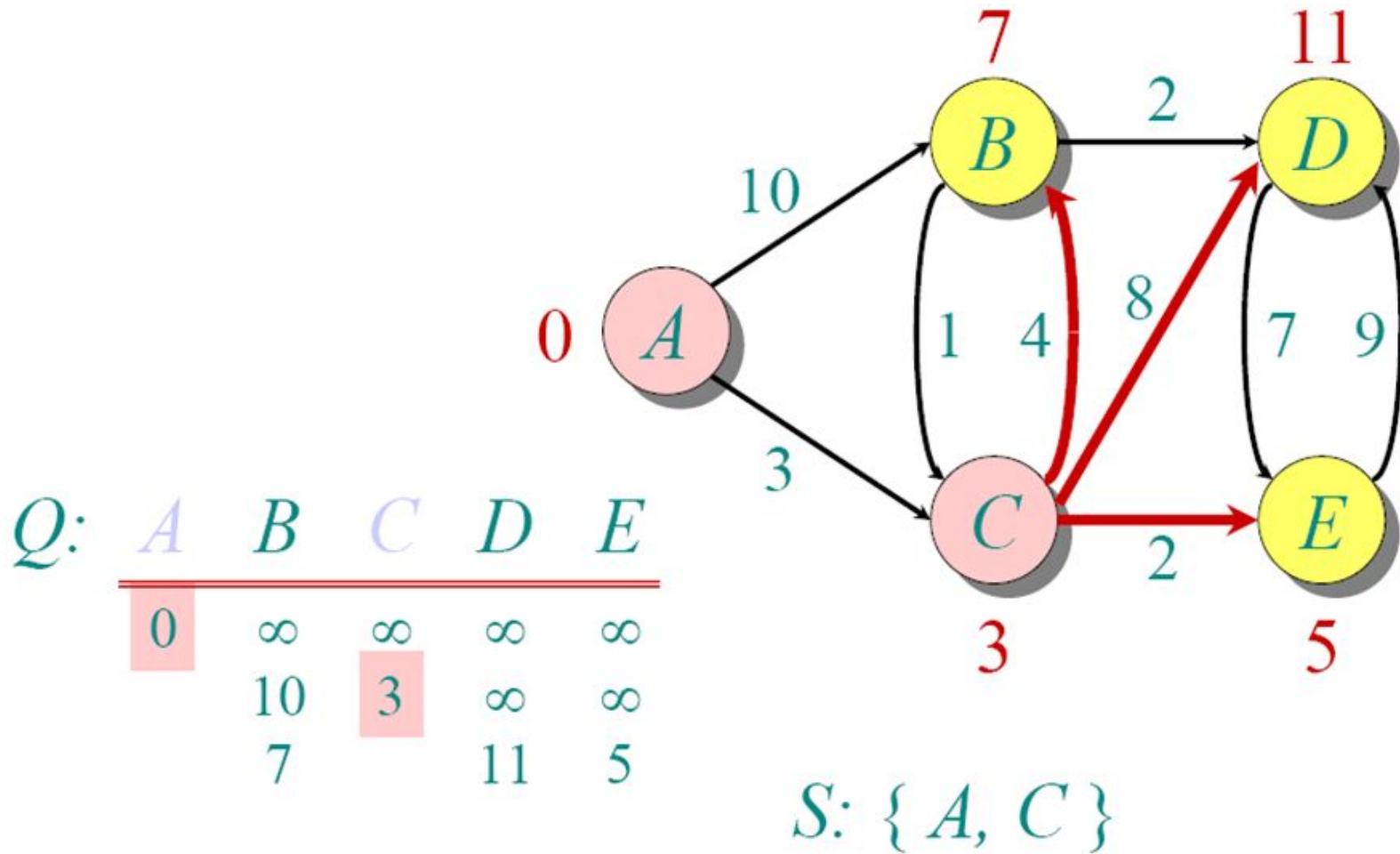
Another Example



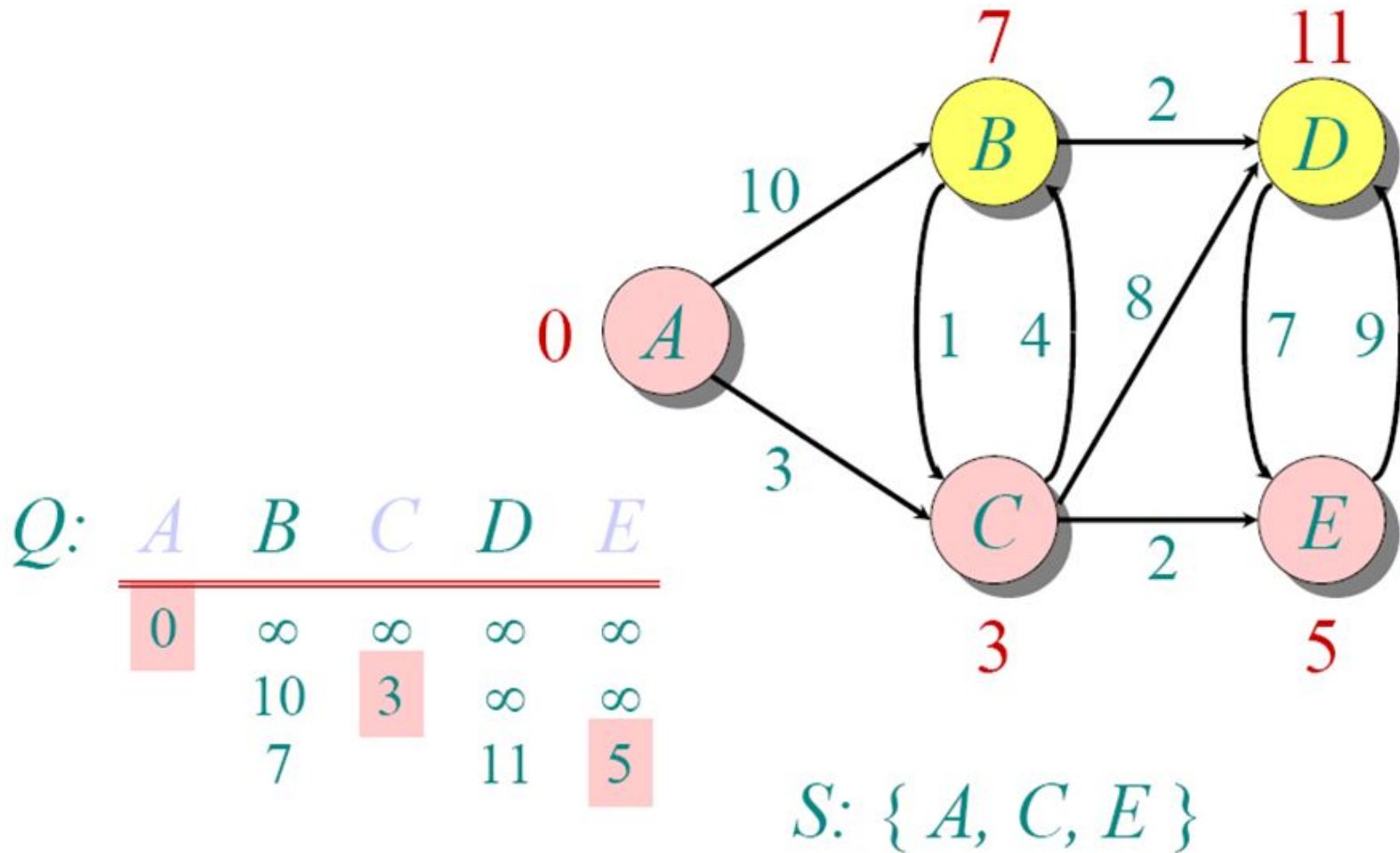
Another Example



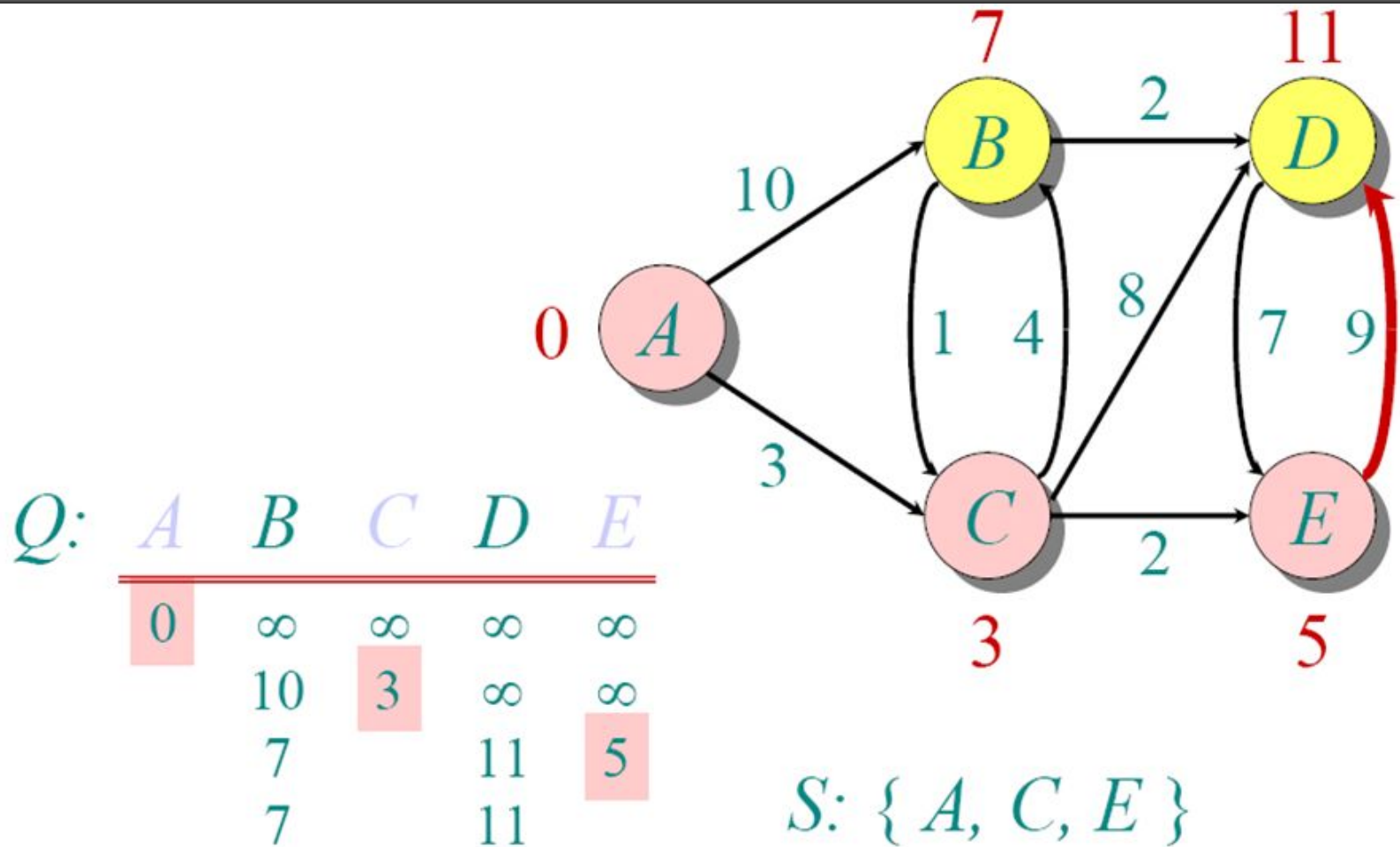
Another Example



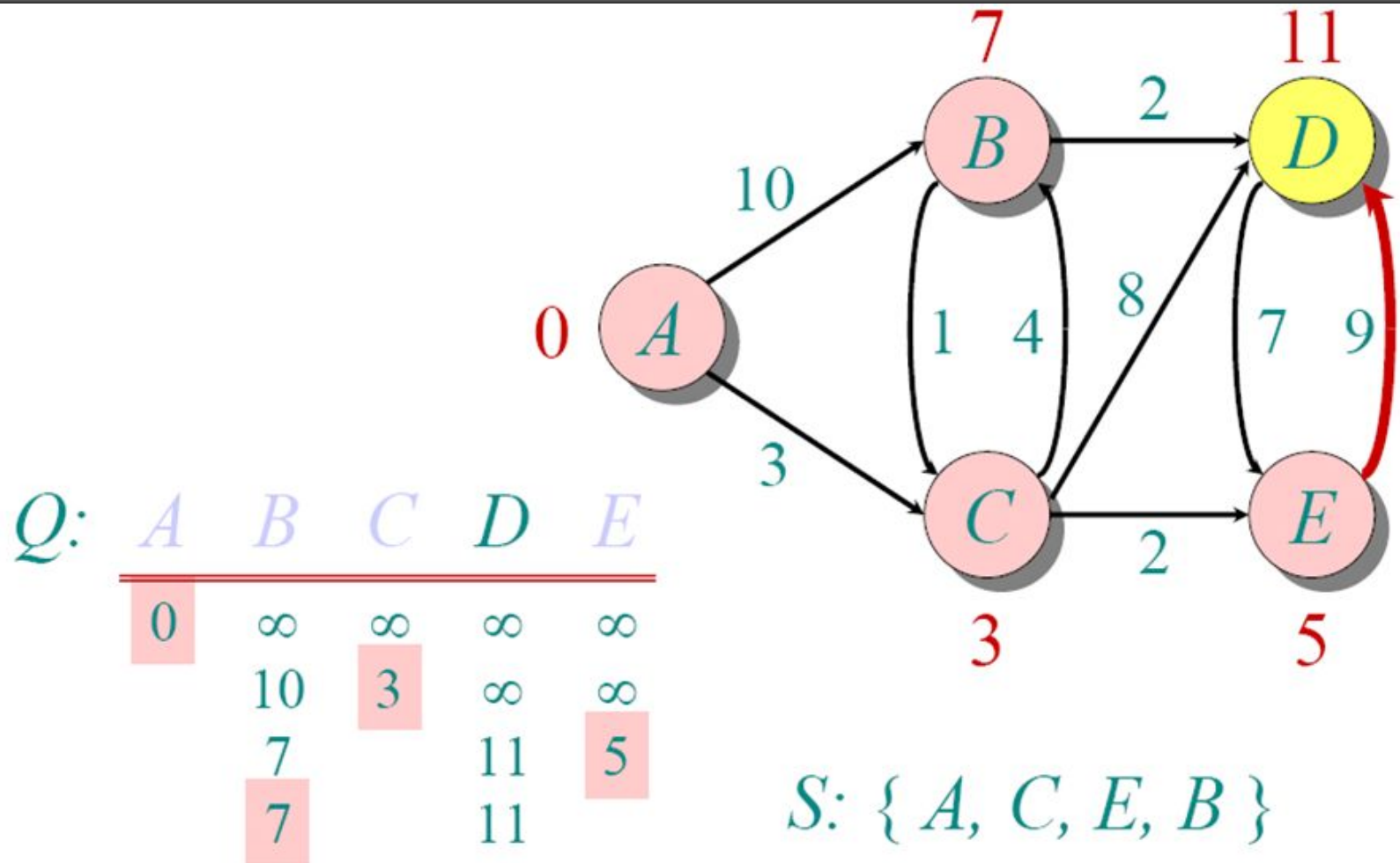
Another Example



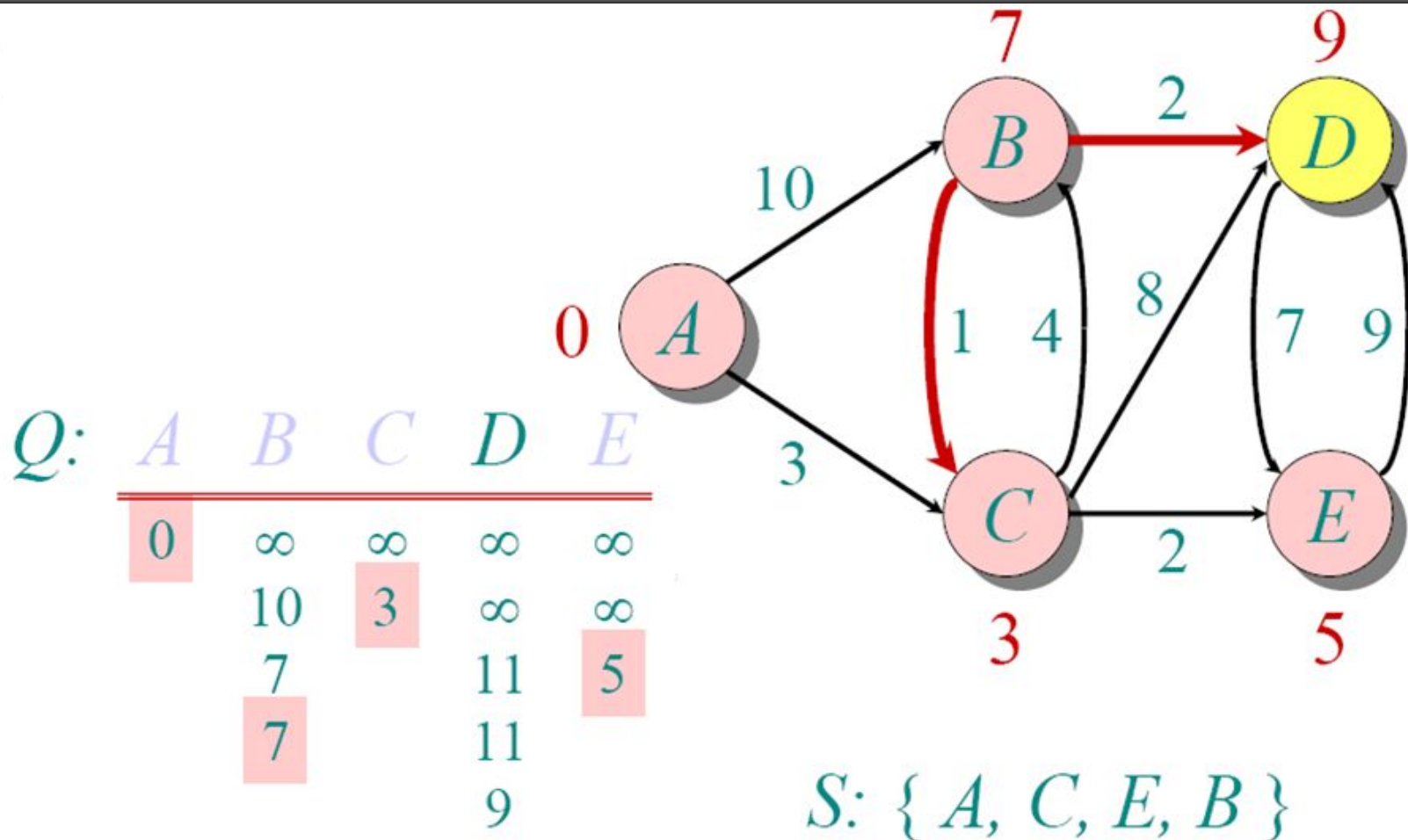
Another Example



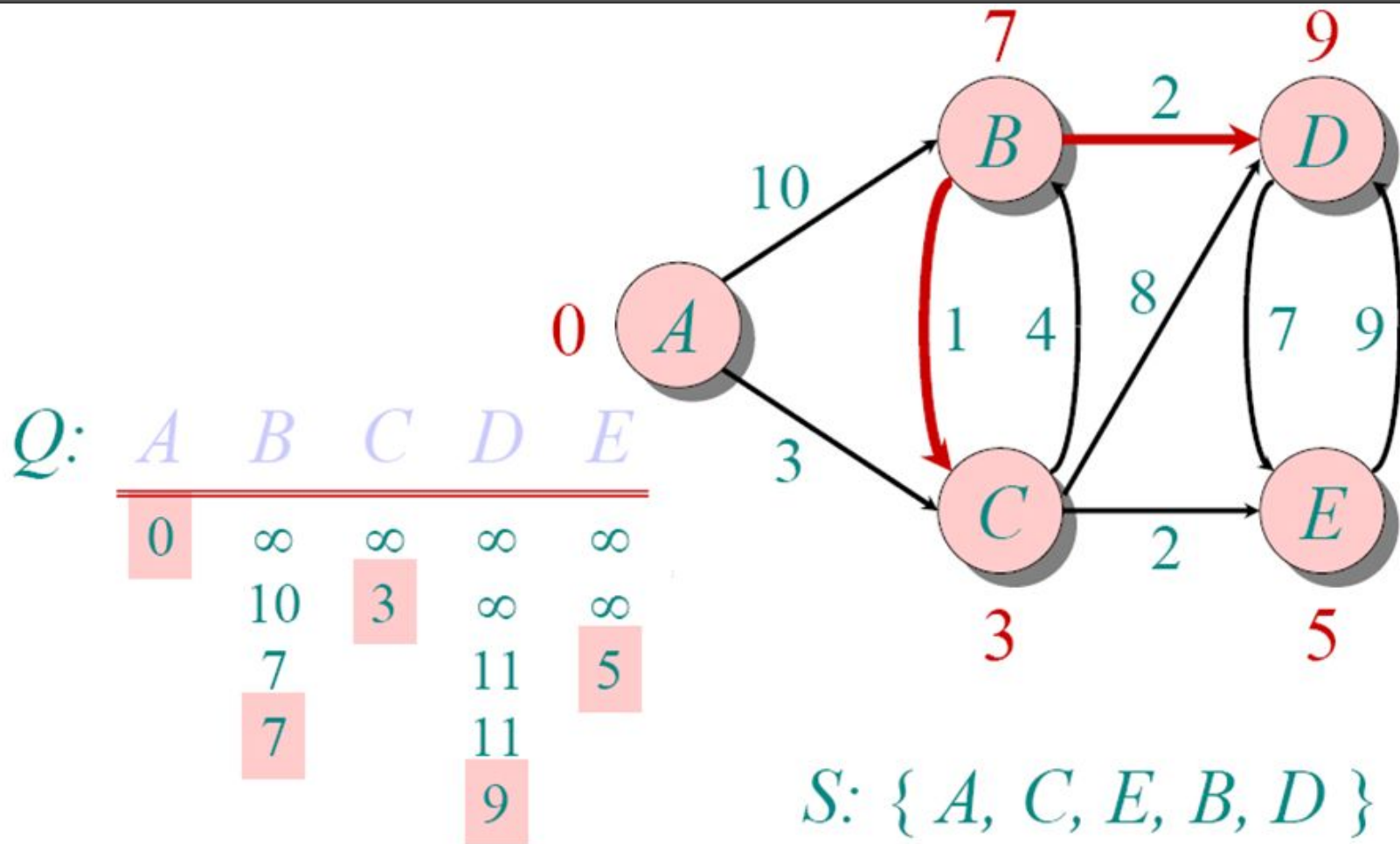
Another Example



Another Example

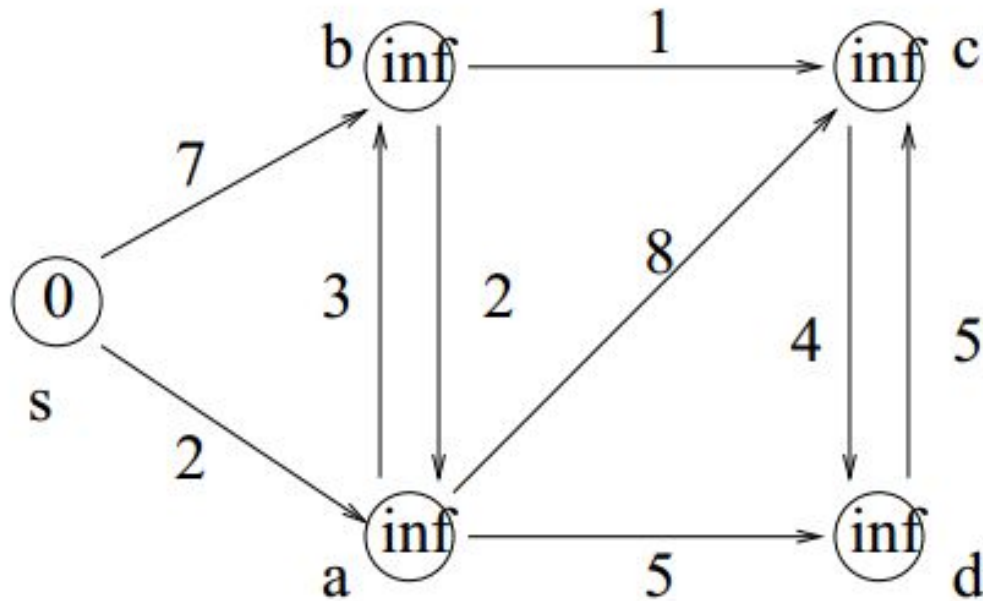


Another Example

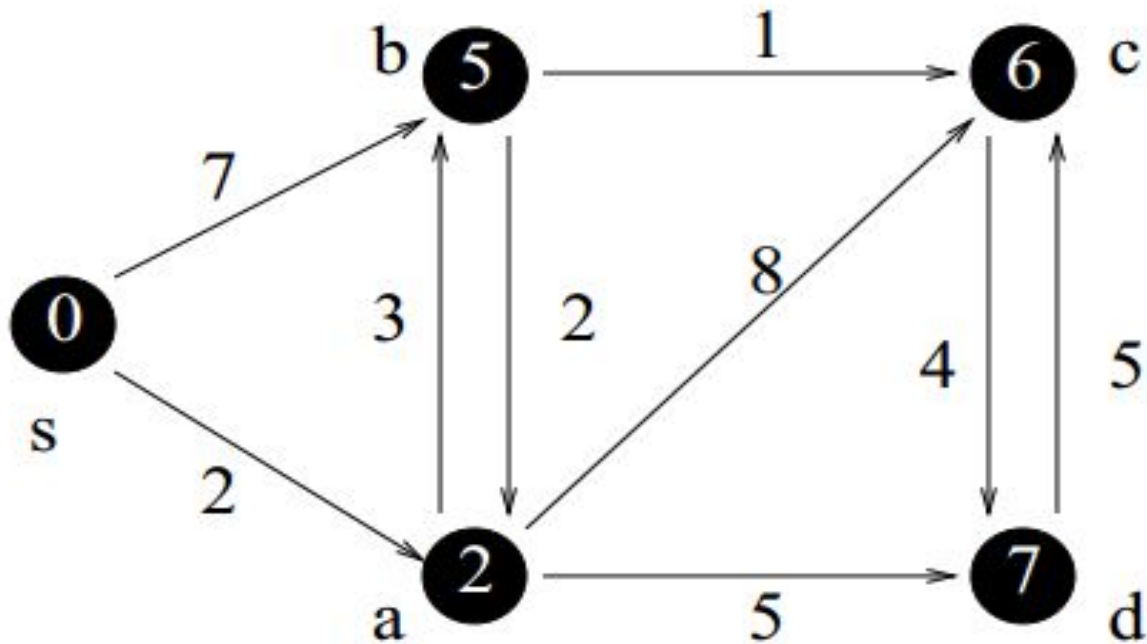


Exercise 1

Find the shortest path to every vertices on this graph

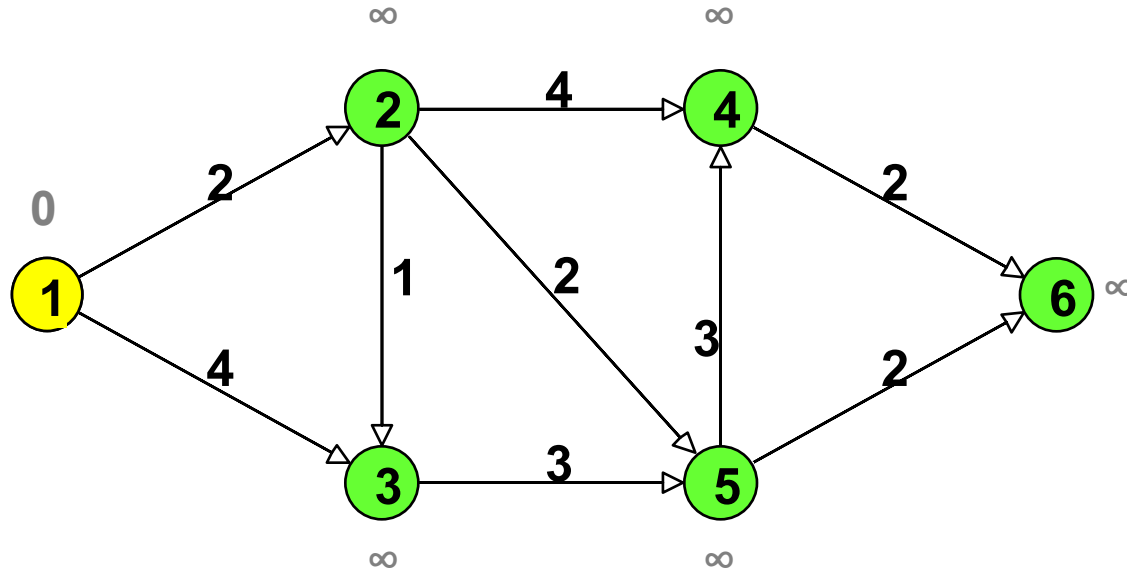


Solution



v	a	b	c	d
	2	5	6	7

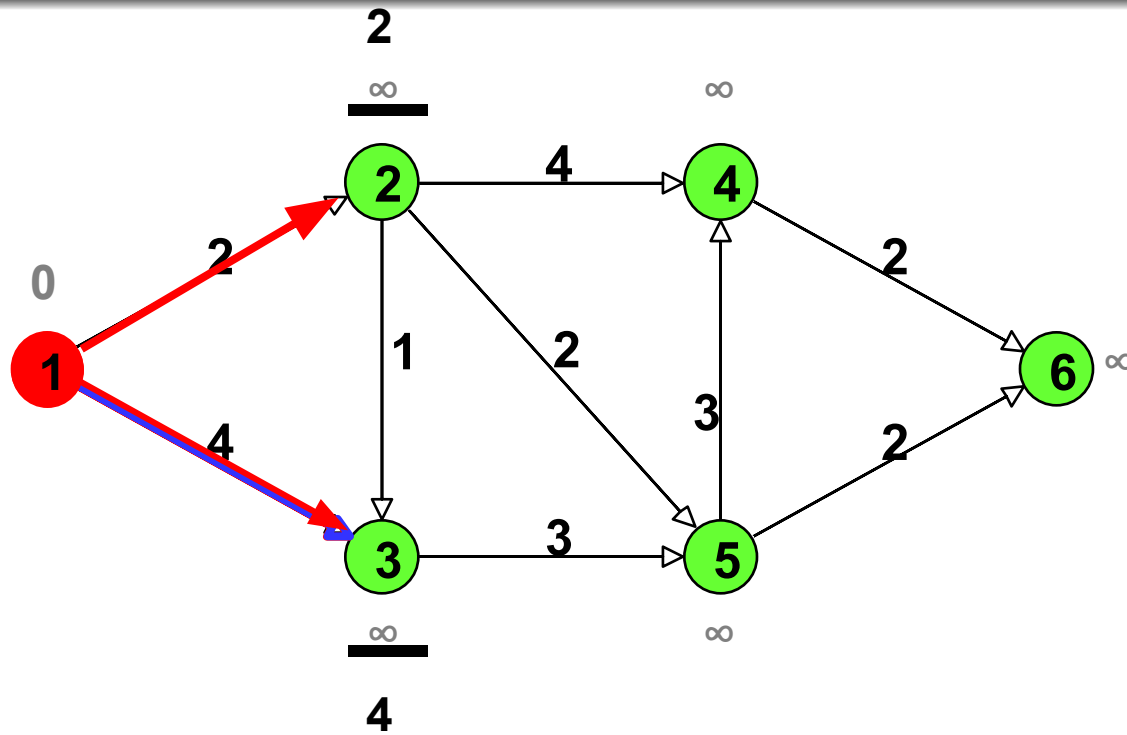
An Example



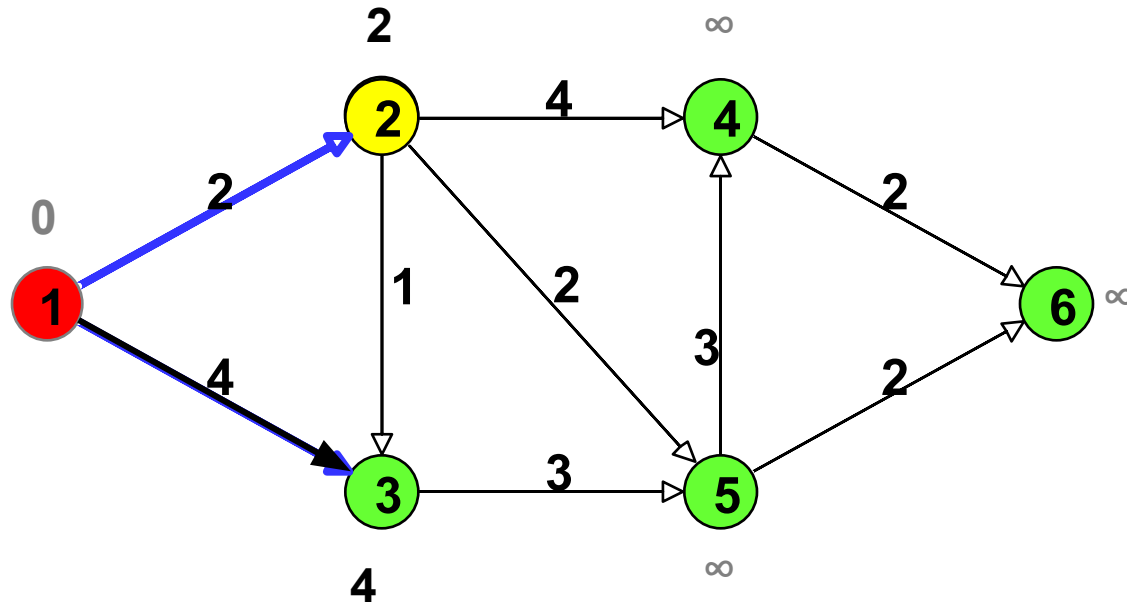
Initialize

Select the node with the minimum temporary distance label.

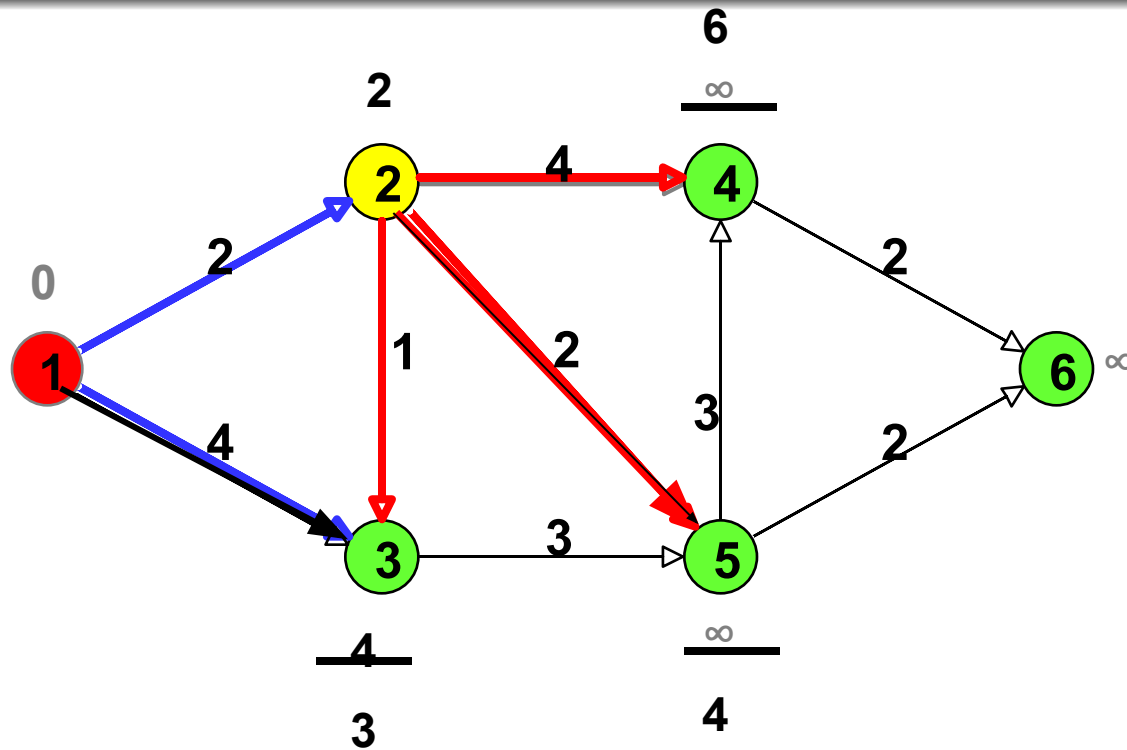
Update Step



Choose Minimum Temporary Label

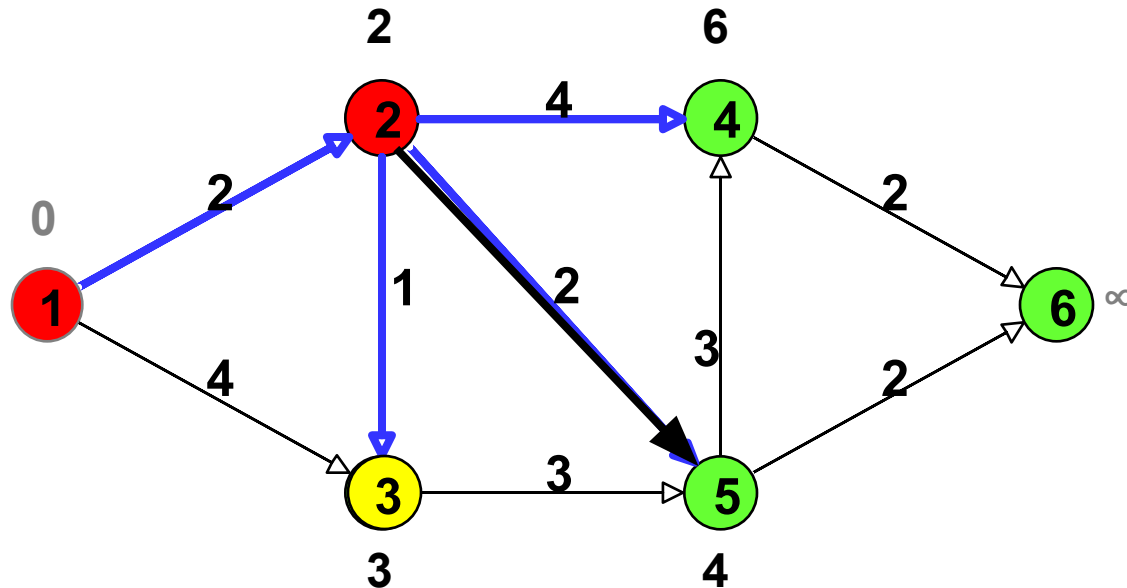


Update Step

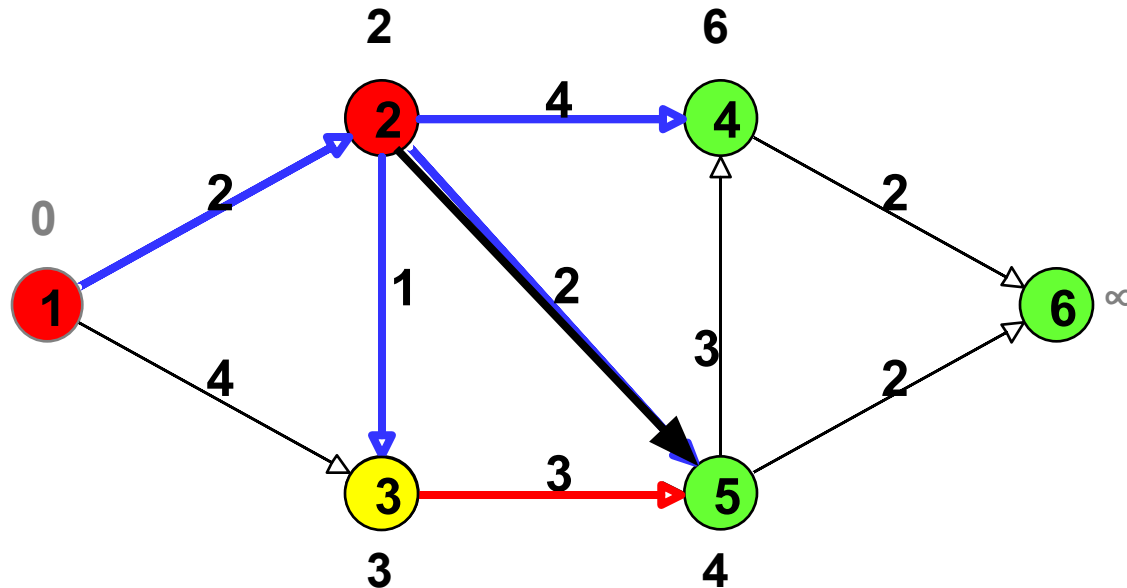


The predecessor of
node 3 is now node 2

Choose Minimum Temporary Label

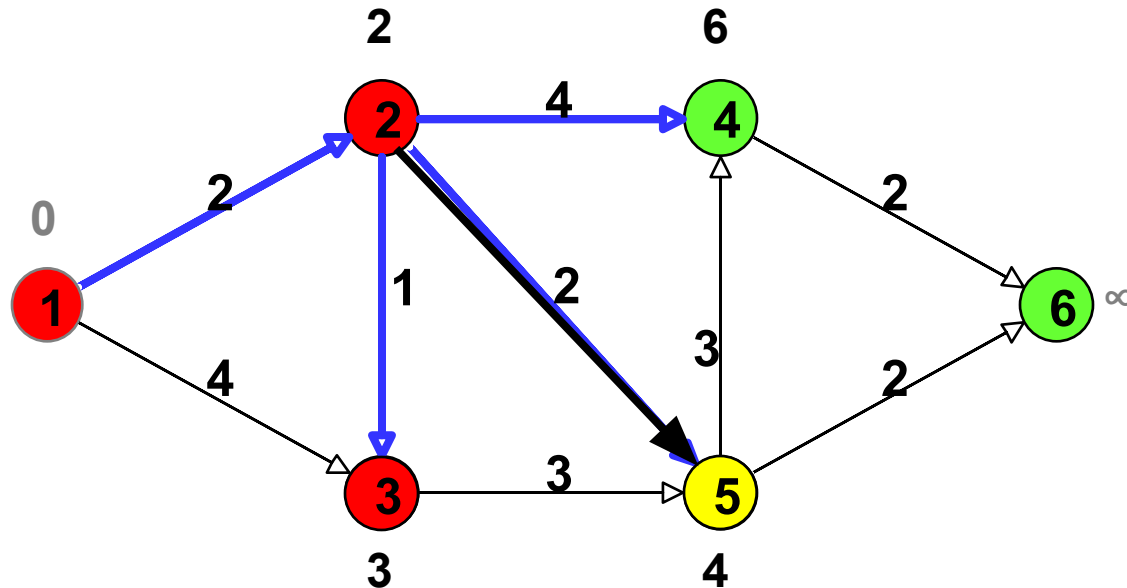


Update

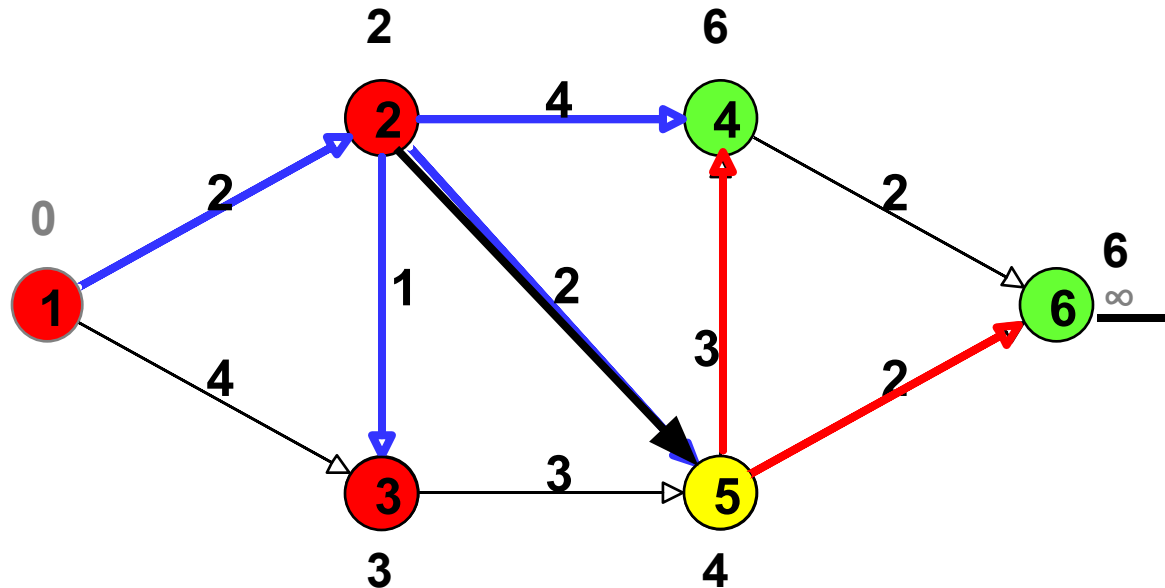


d(5) is not changed.

Choose Minimum Temporary Label

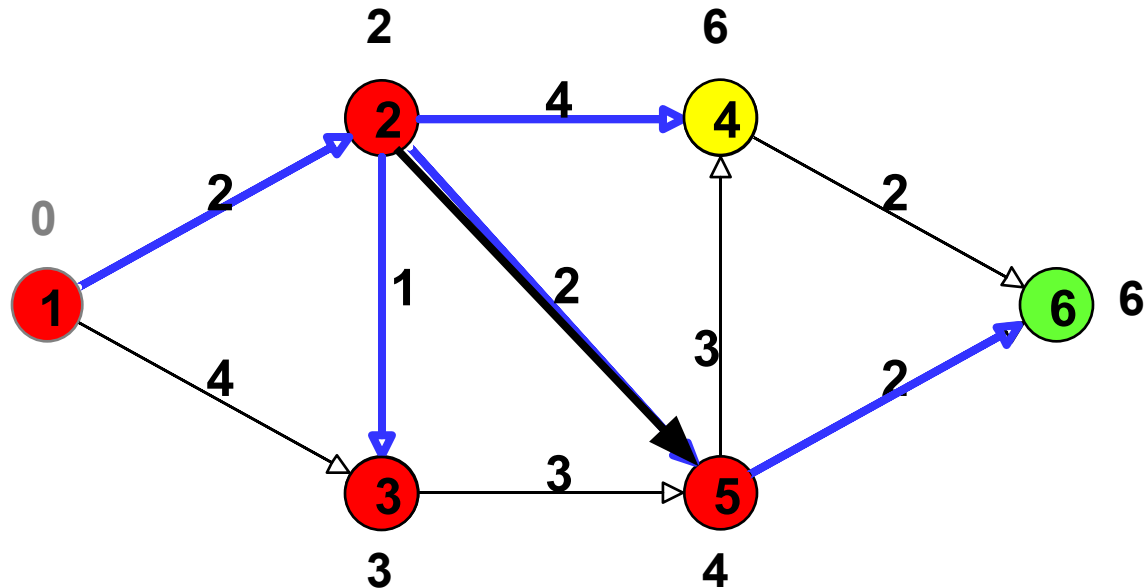


Update

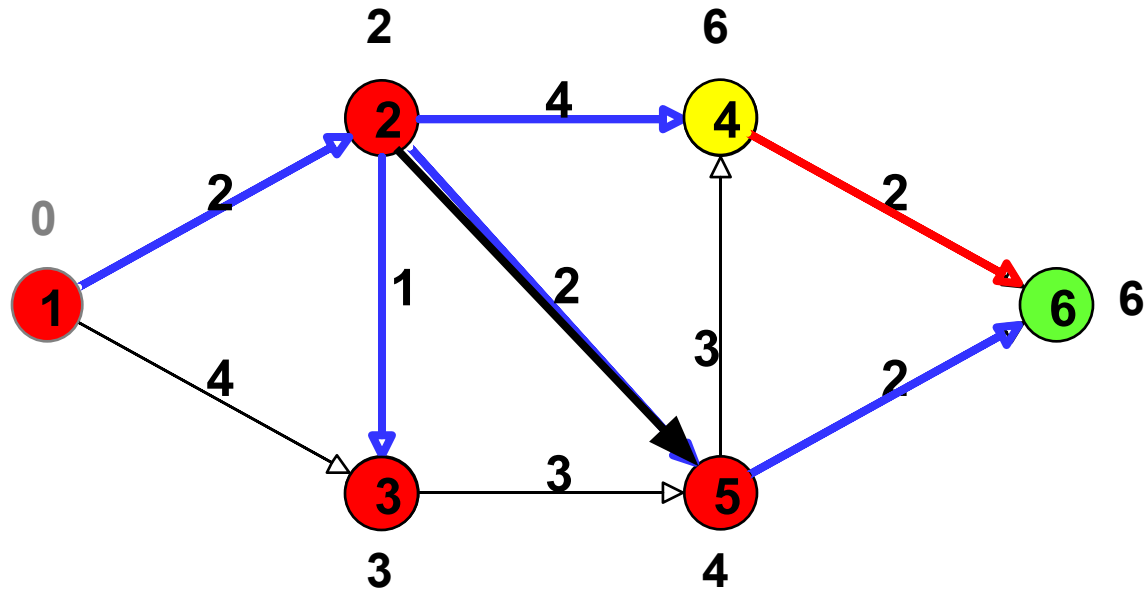


d(4) is not changed

Choose Minimum Temporary Label

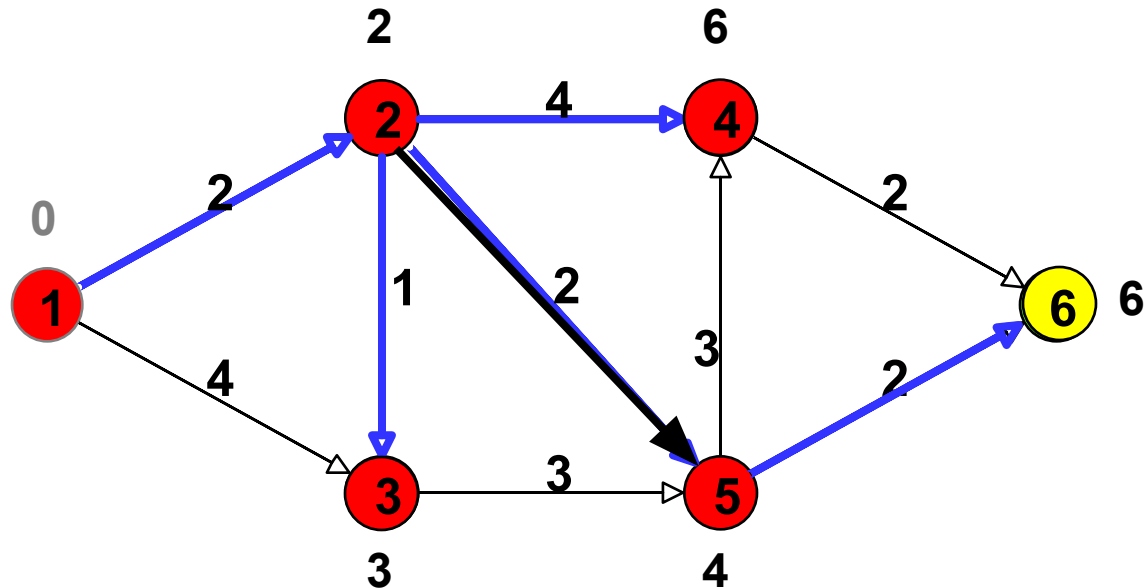


Update



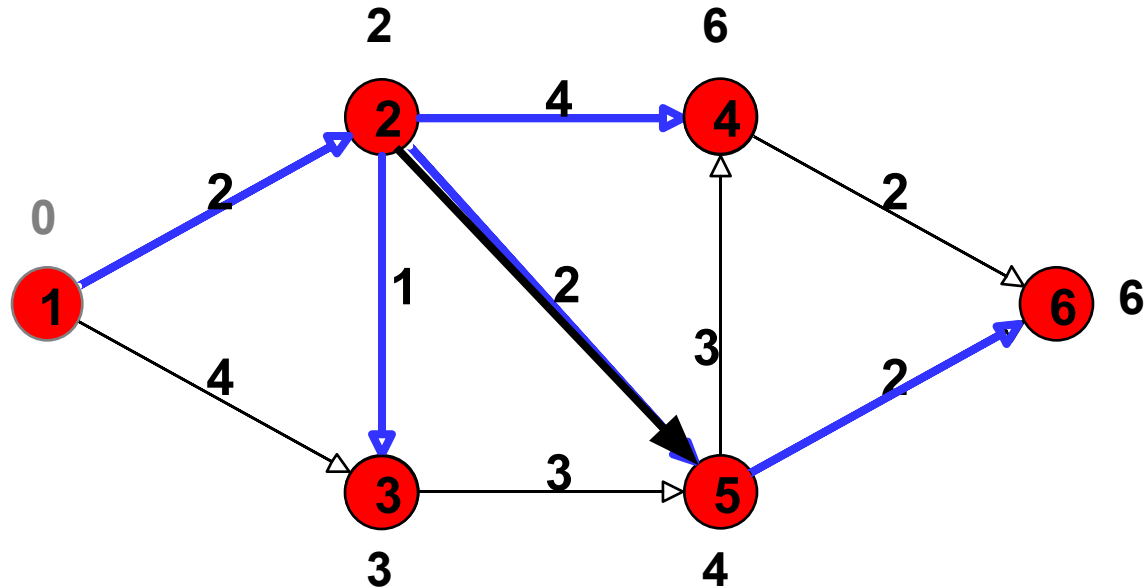
d(6) is not updated

Choose Minimum Temporary Label



There is nothing to update

End of Algorithm



All nodes are now permanent

The predecessors form a tree

The shortest path from node 1 to node 6 can be found by tracing back predecessors

Dijkstra's Pseudo Code

- Graph G , weight function w , root s

DIJKSTRA(G, w, s)

```
1 for each  $v \in V$ 
2     do  $d[v] \leftarrow \infty$ 
3  $d[s] \leftarrow 0$ 
4  $S \leftarrow \emptyset$   $\triangleright$  Set of discovered nodes
5  $Q \leftarrow V$ 
6 while  $Q \neq \emptyset$ 
7     do  $u \leftarrow \text{EXTRACT-MIN}(Q)$ 
8          $S \leftarrow S \cup \{u\}$ 
9         for each  $v \in \text{Adj}[u]$ 
10             do if  $d[v] > d[u] + w(u, v)$ 
11                 then  $d[v] \leftarrow d[u] + w(u, v)$ 
```

relaxing
edges

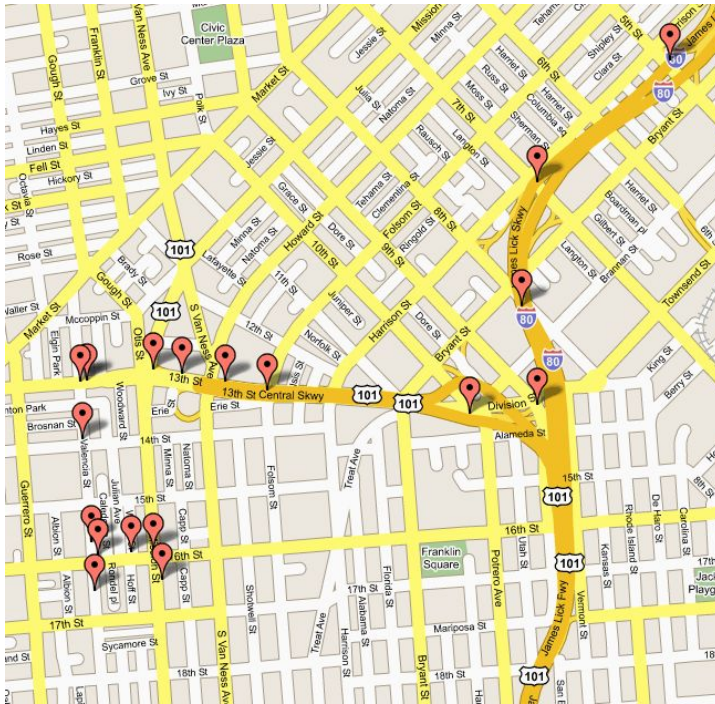
DIJKSTRA'S ALGORITHM - WHY USE IT?

- As mentioned, Dijkstra's algorithm calculates the shortest path to every vertex.
- However, it is about as computationally expensive to calculate the shortest path from vertex u to every vertex using Dijkstra's as it is to calculate the shortest path to some particular vertex v .
- Therefore, anytime we want to know the optimal path to some other vertex from a determined origin, we can use Dijkstra's algorithm.

Applications of Dijkstra's Algorithm

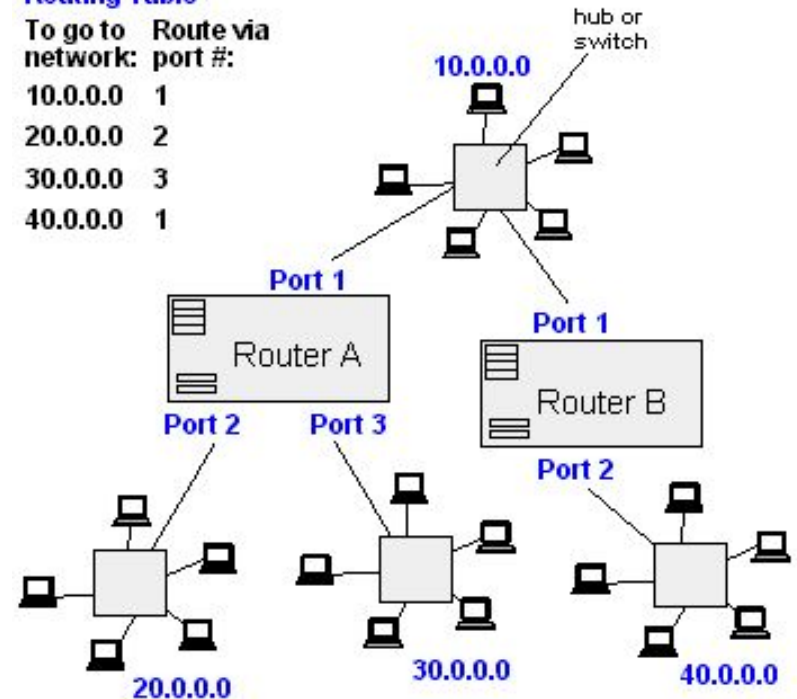
- Traffic Information Systems are most prominent use
- Mapping (Map Quest, Google Maps)
- Routing Systems

From Computer Desktop Encyclopedia
© 1998 The Computer Language Co. Inc.



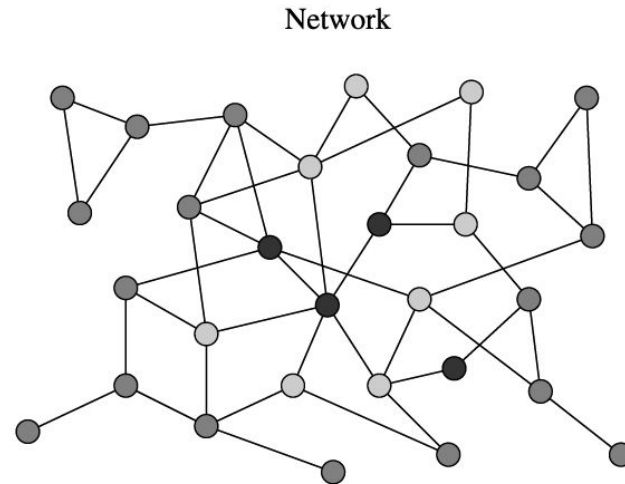
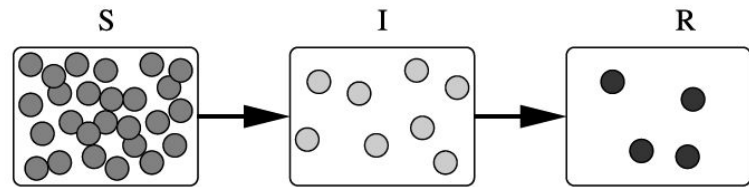
Router A Routing Table

To go to network:	Route via port #:
10.0.0.0	1
20.0.0.0	2
30.0.0.0	3
40.0.0.0	1



Applications of Dijkstra's Algorithm

- One particularly relevant this week: epidemiology
- Prof. Lauren Meyers (Biology Dept.) uses networks to model the spread of infectious diseases and design prevention and response strategies.
- Vertices represent individuals, and edges their possible contacts. It is useful to calculate how a particular individual is connected to others.
- Knowing the shortest path lengths to other individuals can be a relevant indicator of the potential of a particular individual to infect others.



Dijkstra (G, w, s)

```

1.  INITIALIZE-SINGLE-SOURCE( $V, s$ )  $\leftarrow \Theta(V)$ 
2.   $S \leftarrow \emptyset$ 
3.   $Q \leftarrow V[G] \leftarrow O(V)$ 
4.  while  $Q \neq \emptyset$   $\leftarrow$  Executed  $O(V)$  times
5.      do  $u \leftarrow \text{EXTRACT-MIN}(Q) \leftarrow O(\lg V)$ 
6.       $S \leftarrow S \cup \{u\}$ 
7.      for each vertex  $v \in \text{Adj}[u]$   $\leftarrow O(E)$  times
8.          do  $\text{RELAX}(u, v, w)$ 
9.          Update  $Q$  ( $\text{DECREASE\_KEY}$ )  $\leftarrow O(\lg V)$ 

```

$\left. \begin{array}{l} \text{lines 4-6} \\ \text{lines 7-9} \end{array} \right\} O(V \lg V)$
 $\left. \begin{array}{l} \text{lines 7-9} \end{array} \right\} O(E \lg V)$

Running time: $O(V \lg V + E \lg V) = O(E \lg V)$

Question

- Prove that, if there exists negative edge, dijkstra's shortest path algorithm may fail to find the shortest path
- Print the shortest path for dijkstra's algorithm
- How many shortest paths are there from source to destination?
- Suppose you are given a graph where each edge represents the path cost and each vertex has also a cost which represents that, if you select a path using this node, the cost will be added with the path cost. How can it be solved using Dijkstra's algorithm?

Question

- How to solve ACM534 – Frogger?